

AI Log - BTC/Kalshi Dashboard Visualization Design

Representative conversation: designing and revising the Streamlit dashboard pages, charts, and interaction model.

Context

The user wanted a new implementation of the FA550 BTC capstone dashboard. The dashboard needed to visualize how short-dated Kalshi BTC contracts move with respect to BTC futures, while avoiding any writeup about trading strategy. The focus shifted to price behavior, threshold risk, sensitivity, and time-to-close effects.

Visuals requested by user

The user asked to include decision time distribution, contract evolution, threshold-distance heatmaps, pin-risk overlays, scatter plots of short-horizon futures and contract moves, binned sensitivity curves, average contract paths, uncertainty curves, and large futures move event windows. The user also required that the app load at most one week of data at a time.

Dashboard structure developed

The assistant iteratively built a Streamlit app with pages/tabs for Overview, Contract Evolution / Price Behavior, Futures Microstructure, Decision Time, Threshold Risk, Sensitivity, and Data Preview. A sidebar week selector replaced open-ended date selection. The app used contract filters, time-to-close buckets, moneyness buckets, data source period filters, and chart options.

Iterative visualization improvements

Several charts were revised after the user judged them unclear or redundant. The right-side histogram on the overview page was replaced with a more meaningful outcome chart. A broken Average Paths page was removed because time-to-close bucket grouping produced disconnected slivers. On the Contract Evolution page, a duplicate contract-only line chart was replaced with a distribution of 5-second YES price moves and a box plot of absolute moves by time-to-close bucket.

Important UI/debugging issue

The user encountered a Plotly error when adding vertical lines with datetime annotations. The assistant avoided Plotly's datetime annotation issue by using `add_shape` and `add_annotation` instead of `add_vline` for decision and close markers.

```
TypeError: Addition/subtraction of integers and integer-arrays with Timestamp is no longer supported.  
Instead of adding/subtracting `n`, use `n * obj.freq`
```

Outcome

The dashboard became more defensible for a class submission: it has weekly loading, interpretable tabs, price-move distributions, combined contract/underlying evolution, threshold-risk visuals, sensitivity scatter/curve views, decision-time summaries, and a Data Preview section for auditability.